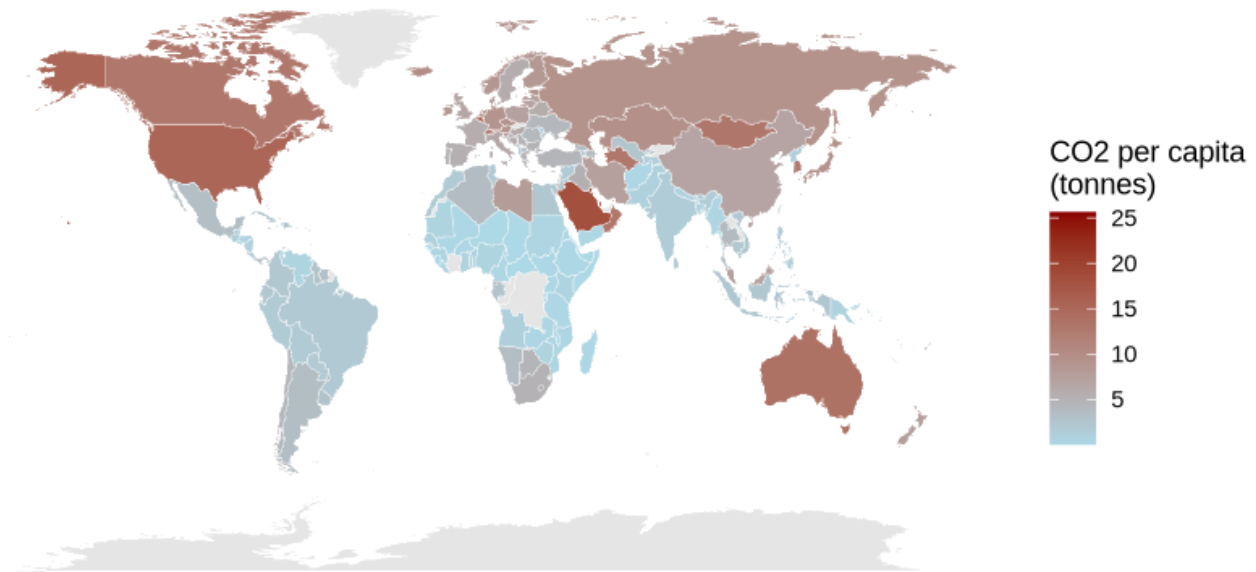
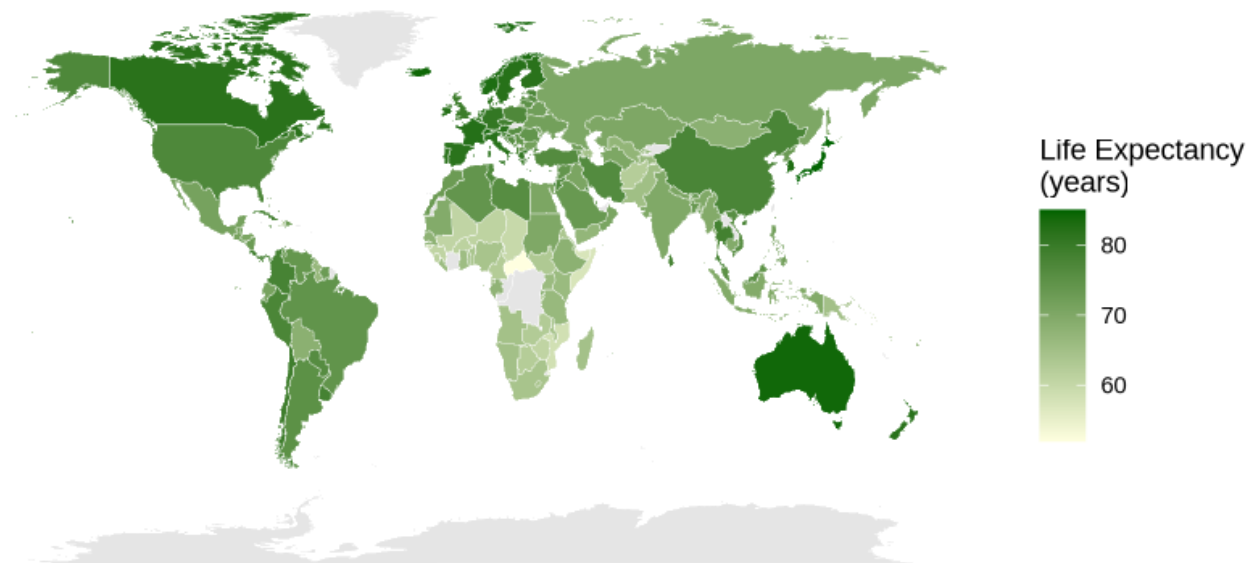


Life expectancy vs CO2 emissions: distinguishing efficient development from low industrialization

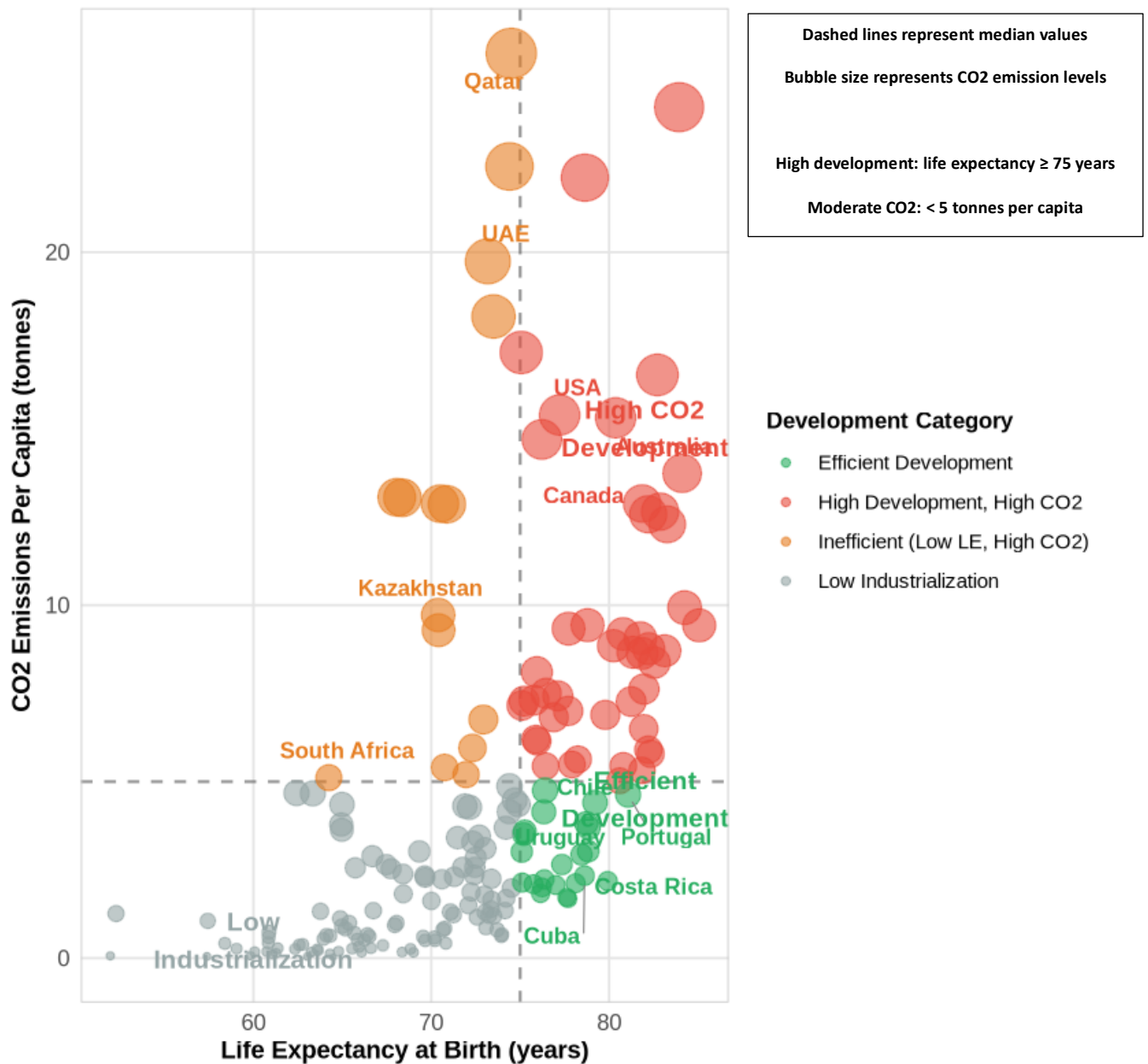
CO2 Emissions Per Capita by Country (2020)



Life Expectancy at Birth by Country (2020)



Life expectancy vs CO2 emissions (2020) – quadrant analysis



This quadrant analysis divides countries into four categories based on median values:

- **High LE, High CO2 (Red):** 70 countries - Developed nations with carbon-intensive economies
- **High LE, Low CO2 (Green):** 26 countries - The "sustainable development" model
- **Low LE, High CO2 (Orange):** 26 countries - Inefficient development pattern
- **Low LE, Low CO2 (Blue):** 69 countries - Developing nations with low emissions

Important caveat: **this correlation does not imply causation.** CO2 emissions don't directly cause longer life expectancy; rather, both are outcomes of economic development and industrialization.

This pattern indicates that both indicators are associated with economic development: wealthier nations tend to have higher energy consumption (and thus CO2 emissions) and better

healthcare/living conditions (higher life expectancy). But the data also suggests that, while development is associated with both higher emissions and better health outcomes, there may be opportunities for countries to achieve high life expectancy without proportionally high emissions.

CONCLUSIONS

1. Efficient development is possible, but rare

Only 23 out of 191 countries (12%) achieve high life expectancy with moderate CO2 emissions. These include both developed nations (Portugal) and middle-income countries (Costa Rica, Chile, Uruguay).

2. Low industrialization vs sustainability

107 countries (56%) have low CO2 simply because they lack economic and industrial development. These countries also have lower life expectancy (<75 years), indicating they haven't achieved full development yet.

3. The "efficient development" model

Countries like Costa Rica, Portugal, Chile, and Uruguay demonstrate that development doesn't require extreme carbon intensity. These nations have achieved life expectancy comparable to high-emission countries (75-81 years) while emitting 1.7-4.8 tonnes per capita - compared to USA (78.5 years, 13.0 tonnes) or Australia (83.2 years, 14.6 tonnes).

4. The challenge

Most developed countries (46 nations) still follow the high-CO2 pathway. 15 countries show the worst pattern: relatively low life expectancy despite high emissions (inefficient development)

5. Policy implications

The efficient development countries prove that quality healthcare, education, and social systems can deliver long life expectancy without proportionally high emissions. Factors enabling this include renewable energy, efficient public transport, universal healthcare systems, and less car-dependent urban planning.

MY REFLECTIONS

In my opinion, the interaction with Julius only seemed easy. While it's true that you can describe what you want in natural language, you still need (at least) some data skills to judge whether the results are realistic or meaningful—and to choose the best way to present them. Julius helped me a lot with tasks, code (I had no experience at all!), and ideas, but without "supervision," conclusions could easily turn into... misconclusions! I need to study... MORE!